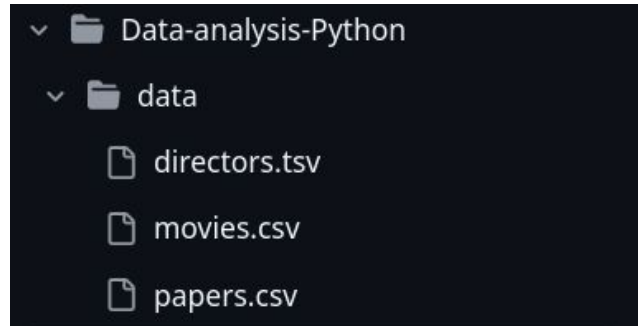


Data analysis and vizualization in Python

Data visualization in Python - RSE course

Tools and data

- libraries
 - Pandas - for statistics using tabular data
 - matplotlib - data vizualization
- datasets in this course
 - data as dictionary
 - csv - organism.csv, papers.csv
 - tsv - habitats.tsv



Pandas

- DataFrame = table like structure
- fundamental methods of dataframes - `.head()`, `.info()`, `.describe()`
- enables filtering, modifying and aggregating data
- loading datasets into dataframe:
 - CSV - `pd.read_csv()`
 - JSON - `pd.read_json()`
 - EXCEL - `pd.read_excel()`

DataFrame

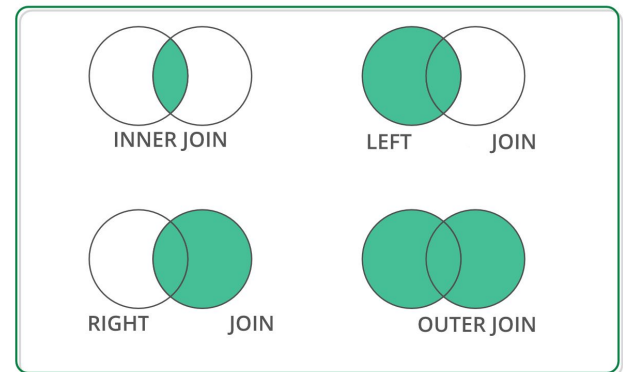
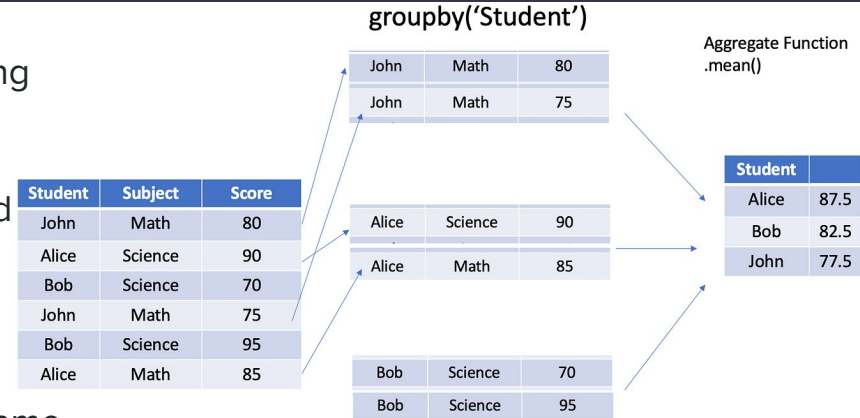
	Name	Age	Sex	label
0				row
1				
2				
3				
index		column		

Selecting, filtering and modifying

- selecting - `dataframe["column_name"]`
- filtering - conditions to filter rows where it is True
 - `<, >, ==, !=` : `dataframe[dataframe["column_name"] > 3]`
 - `&, |, ~` combining conditions: `dataframe[(dataframe["column_name"]=="str") | (df["column_name"]=="str1")]`
- modifying
 - `dataframe["column_name"] = dataframe["column_name"] * 10`
 - `dataframe.rename(columns={"column_name": "new_column_name"})`
- drop - removing specific column

Aggregation

- **groupby**("column_name") - organising data by grouping rows from one column that share common value
- after that we can focus on another specific column and provide some info:
 - **.mean(), .count(), sum(), min(), max(), median()**
 - `dataframe.groupby("column_name")["column_name 2"].count()`
- **merge** - two dataframes with column with same keys
 - `merge(dataframe1, dataframe2, on="name_of_column_name", how="left")`

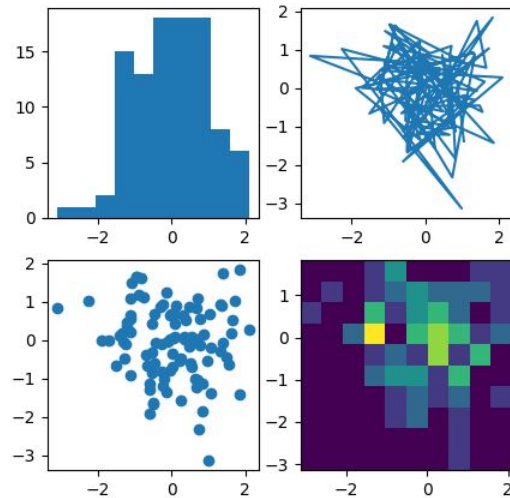


Errors in data

- pandas recognize NaN, nan, NAN, NULL, null, None, NA, N/A, ?,... and converts it to **NaN**
- how to detect issues?
 - `dataframe.info()`
 - `dataframe.describe()`
 - `dataframe.isna()` / `dataframe["column_name"].isna()` - **finds if** there are missing data
 - `dataframe.duplicated()` - returns True for duplicate rows
- how to fix it?
 - `dataframe["column_name"].fillna(0)` - **replaces** missing values with anything we want (e.g. 0)
 - we can also use mean - `df["salary"].fillna(df["salary"].mean())`
 - `dataframe.dropna()` - **removes** rows with missing values
 - `dataframe["column_name"].astype(int)` - **converts** column to **correct type**
 - `dataframe.drop_duplicates()` - removes duplicate rows

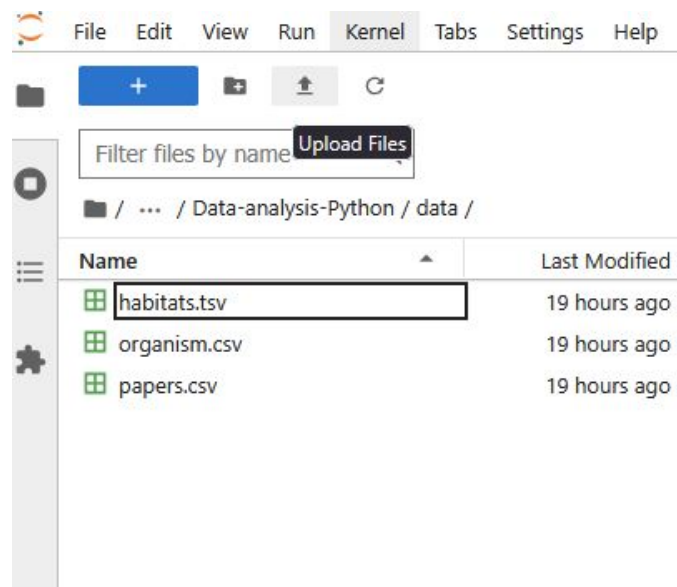
Matplotlib

- standard Python library for data visualisation
- line plots, scatter plots, histograms, boxplots, heatmaps
- plot customization - labels, titles, size, colors...
- comparison of multiple datasets
- **fig** = the **canvas** (the whole figure / drawing area)
- **axes** = the **plot inside the canvas** (where the actual data is drawn)
- **plt** = the **director** that keeps track of everything and decides what gets displayed



Mini project

- dataset papers.csv
- if you have some csv, tsv or excel type of dataset that is yours use it!!
 - save it into /data folder



Key takeaways

- datasets are loaded in **data frames**
 - data frames can be **modified** or **filtered**
 - **error data** needs to be recognized and **fixed**
 - easy to compute **statistics**
 - **Matplotlib** is library for **visualization**
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